Strong triangularity of the Hermitian Jacobi operator

MathDB 376317: a Casimir proof

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Task 1: the original problem. We prove Kuijper's Conjecture 3.13 [1], with strong triangularity understood exactly as in Definition 2.12 and with the ordering on pp. 19–20. The diagonal coefficient is independent of every choice within a multiplicity space.

Theorem 1. *Let $m, k \geq 1$ and $\kappa_j > m/2 - 1$ for $1 \leq j \leq k+1$. Put*

$$N = \sum_{j=1}^{k+1} \kappa_j + \frac{(k+1)m}{2}.$$

For every multivariate Schur basis element $s_\theta = s_\phi^{\tau_1,\dots,\tau_k}$, including any allowed multiplicity label, the operator of Definition 3.5 satisfies

$$\mathcal{G}_\kappa^m s_\theta = -\left(\sum_{i=1}^{m} \phi_i(\phi_i + N + 1 - 2i) \right) s_\theta + \sum_{\eta<\theta} a_{\theta\eta} s_\eta. \tag{1}$$

Consequently it is strongly triangular for every choice of the multivariate Schur basis within the specified multiplicity spaces.

An identity for the entire differential operator

Complexify the real vector space of Hermitian matrix tuples and write $x_{j,ab}$ for the entries of X_j. Thus all km^2 entries are independent polynomial coordinates for the calculation; restriction to the Hermitian space recovers the original operator. Write $\partial_{j,ab} = \partial/\partial x_{j,ab}$ and define

$$\mathcal{E} = \sum_{j,a,b} x_{j,ab}\partial_{j,ab}, \qquad R_{ab} = \sum_{j,c} x_{j,ca}\partial_{j,cb}, \qquad L_{ab} = \sum_{j,c} x_{j,ac}\partial_{j,bc}.$$

The R_{ab} and L_{ab} are the column and row actions of $\mathfrak{gl}_m$, respectively. They commute with one another, and each family satisfies the standard matrix-unit commutation relations. Set

$$\mathcal{C}_R = \sum_{a,b} R_{ab}R_{ba}, \qquad \mathcal{C}_L = \sum_{a,b} L_{ab}L_{ba}.$$

The product rule, including differentiation of the coefficient in the inner vector field, gives

$$\mathcal{C}_R = m\mathcal{E} + \sum_{j,\ell,a,b,c,d} x_{j,ad}x_{\ell,cb}\partial_{j,ab}\partial_{\ell,cd}, \tag{2}$$

$$\mathcal{C}_L = m\mathcal{E} + \sum_{j,\ell,a,b,c,d} x_{j,cb}x_{\ell,ad}\partial_{j,ab}\partial_{\ell,cd}. \tag{3}$$

For instance, in $R_{ab}R_{ba}$, differentiation of $x_{\ell,db}$ contributes $\sum_{j,c} x_{j,ca}\partial_{j,ca}$, and summing over a, b gives $m\mathcal{E}$.

Define the operators that decrease degree by one,

$$K_j = \sum_{a,b,c,d} \delta_{ad} x_{j,cb}\partial_{j,ab}\partial_{j,cd} + \Big(\kappa_j + \frac{m}{2}\Big)\sum_a \partial_{j,aa}. \tag{4}$$

The source operator consists of the within-matrix second-order terms $(\delta_{ad} - x_{j,ad})x_{j,cb}\partial_{j,ab}\partial_{j,cd}$, the mixed terms

$$-\frac{1}{2}\sum_{j\neq\ell,a,b,c,d} (x_{j,ad}x_{\ell,cb} + x_{j,cb}x_{\ell,ad})\partial_{j,ab}\partial_{\ell,cd},$$

and the drift $\sum_{j,a,b}((\kappa_j + m/2)\delta_{ab} - Nx_{j,ab})\partial_{j,ab}$. The two quadratic coefficients coincide when $j = \ell$. Combining them with (2)–(3) therefore proves the exact identity, on the full polynomial ring,

$$\boxed{\mathcal{G}_\kappa^m = -\frac{1}{2}(\mathcal{C}_R + \mathcal{C}_L) - (N-m)\mathcal{E} + \sum_{j=1}^k K_j.} \tag{5}$$

Both Casimirs are retained here: they need not agree on noninvariant polynomials. They have the same scalar action on the invariant components used below.

The Casimir scalar, including multiplicities

For either family of matrix units T_{ab}, the element $\sum_{a,b} T_{ab}T_{ba}$ is central; this follows by substituting $[T_{ab}, T_{cd}] = \delta_{bc}T_{ad} - \delta_{ad}T_{cb}$ and cancelling terms in its commutator with T_{cd}. On the irreducible polynomial representation of highest weight $\phi = (\phi_1, \ldots, \phi_m)$ it acts by

$$c_m(\phi) = \sum_{i=1}^m \phi_i(\phi_i + m + 1 - 2i). \tag{6}$$

To verify the normalization, take a highest-weight vector v, so that $T_{ab}v = 0$ for $a < b$ and $T_{aa}v = \phi_a v$. Then

$$\sum_{a,b} T_{ab}T_{ba}v = \left(\sum_a \phi_a^2 + \sum_{a<b}(\phi_a - \phi_b)\right) v = c_m(\phi)v.$$

Centrality extends this equality to the whole irreducible representation.

Recall the Cauchy and tensor-product decompositions defining the source's multivariate Schur spaces (pp. 7 and 16–17). For fixed individual partitions $\boldsymbol{\tau} = (\tau_1, \ldots, \tau_k)$, each matrix contributes a row and a column Schur factor of type τ_j. Decomposing the tensor product of row factors and the tensor product of column factors under their respective diagonal GL_m actions gives types ϕ and ψ. Invariance under simultaneous unitary conjugation selects equal types $\phi = \psi$. This is precisely the construction of the spaces $S_\phi^{\boldsymbol{\tau}}$ in the source.

If ϕ has tensor-product multiplicity d, the corresponding invariant space can have dimension d^2. Nevertheless *each* copy of the row representation has Casimir scalar $c_m(\phi)$, and the same is true for each column copy. A Casimir belongs to the acting enveloping algebra and acts as this scalar times the identity on the entire isotypic component, including its multiplicity factor. Thus

$$\mathcal{C}_R s = c_m(\phi)s, \qquad \mathcal{C}_L s = c_m(\phi)s \quad \text{for every } s \in S_\phi^{\boldsymbol{\tau}}. \tag{7}$$

This asserts more than preservation of an isotypic component by an arbitrary intertwiner; no mixing of equivalent copies is possible. Also $\mathcal{E}s = |\phi|s$, because $|\phi| = \sum_j |\tau_j|$. The degree-preserving part of (5) therefore acts on this entire space as

$$-c_m(\phi) - (N-m)|\phi| = -\sum_{i=1}^{m} \phi_i(\phi_i + N + 1 - 2i).$$

Every remaining term is strictly lower

It remains to check the exact source ordering, not merely total degree. The operator K_j acts only on the entries of X_j, decreases their degree by one, and is invariant under simultaneous unitary conjugation. The last assertion follows directly from the contracted indices in (4), or by conjugating its matrix coordinates by a unitary matrix.

Let W_τ denote the single-matrix Cauchy summand with individual partition τ. A basis element s_θ belongs to $\bigotimes_{\ell=1}^{k} W_{\tau_\ell}$. If $|\tau_j| = 0$, then $K_j s_\theta = 0$. Otherwise

$$K_j s_\theta \in \left(\bigotimes_{\ell \neq j} W_{\tau_\ell} \right) \otimes \mathcal{P}_{|\tau_j|-1}(\mathrm{Mat}_m),$$

with factors placed in their original order. Decomposing the last polynomial space into Cauchy summands shows that every output index $\eta = (\rho_1, \ldots, \rho_k; \psi)$ satisfies

$$\rho_\ell = \tau_\ell \ (\ell \neq j), \qquad |\rho_j| = |\tau_j| - 1.$$

Because the source uses *graded* dominance for each partition, $\rho_j < \tau_j$, regardless of its shape. Since all the other individual partitions agree, the source's reverse lexicographic combination gives $\eta < \theta$, regardless of ψ or its multiplicity label. The invariant output expands in the multivariate Schur basis, so this accounts for every term in $K_j s_\theta$.

Combining this observation with (5) and (7) proves (1). The diagonal coefficient depends only on ϕ, and in particular is identical on equivalent indices. This proves strong triangularity in Definition 2.12 and completes the proof of Theorem 1.

Scope and verification. The argument is algebraic and establishes all matrix sizes, all numbers of variables, all partition degrees, and all multiplicities. In fact, the operator identity and triangularity hold for arbitrary complex parameters; the theorem states the parameter range of the question. The companion exact checker tests the operator identity on a basis of quadratic polynomials in a nontrivial matrix case and verifies a four-dimensional repeated-type invariant space. These tests check conventions; the proof for arbitrary dimensions is above. The review was performed by the authoring agent, without independent peer review. This theorem is separate from the product-eigenfunction assertion of Conjecture 3.12.

Four follow-up tasks

Task 2. Compute the lower coefficients $a_{\theta\eta}$ explicitly in a fixed multivariate Schur basis, starting with the first tensor-product multiplicity greater than one, and express the general coefficients through branching maps.

Task 3. Construct a commuting family of symmetric differential operators that separates the partition and multiplicity labels left indistinguishable by the single quadratic Casimir.

Task 4. Classify coincidences between the values $\sum_i \phi_i(\phi_i + N + 1 - 2i)$ for admissible N and determine how they affect canonical choices of orthogonal eigenpolynomials.

Task 5. Use tensor product multiplicities to compute a generating function for the multiplicities in the Jacobi spectrum. Derive corresponding heat-trace asymptotics in the geometric parameter range.

References

[1] T. Kuijper, *Jacobi polynomials on the simplex of Hermitian matrices and zonal spherical functions on partial flag manifolds*, arXiv:2607.25760v1 (2026), Definitions 2.5, 2.12, and 3.5, and Conjecture 3.13.